

Solutions: R Programming Questions

Unit 2: Empirical Study of Sampling Distributions

1. Probability and Probability Distributions:

Solution: To generate 1000 random numbers from a normal distribution with mean 10 and standard deviation 3:

```
set.seed(123)
```

```
random_data <- rnorm(1000, mean = 10, sd = 3)
```

```
hist(random_data, main = "Normal Distribution (mean=10, sd=3)")
```

2. Generating Random Numbers:

Solution: To generate 20 random numbers from a uniform distribution between 5 and 15:

```
set.seed(123)
```

```
random_uniform <- runif(20, min = 5, max = 15)
```

```
mean(random_uniform)
```

```
sd(random_uniform)
```

3. Selecting Random Samples:

Solution: To select a random sample of size 10 from a vector of numbers 1 to 50 and repeat this 5 times:

```
set.seed(123)
```

```
vector_data <- 1:50
```

```
samples <- replicate(5, sample(vector_data, 10))
```

```
apply(samples, 2, mean)
```

4. Empirical Study of the Sampling Distribution of Estimators:

Solution: To generate 1000 random samples of size 50 from a normal distribution with mean 20 and standard deviation 5 and calculate the sample means:

```
set.seed(123)
```

```
sample_means <- replicate(1000, mean(rnorm(50, mean = 20, sd = 5)))
```

```
hist(sample_means, main = "Sampling Distribution of Sample Means", xlab = "Sample Means")
```

Unit 3: Data Simulation

1. Simulation of Data from a Probability Distribution:

Solution: To simulate 500 data points from an exponential distribution with rate parameter 0.5:

```
set.seed(123)
```

```
simulated_exp_data <- rexp(500, rate = 0.5)
```

```
hist(simulated_exp_data, main = "Exponential Distribution (rate = 0.5)")
```

2. Simulation of Data for a Regression Model:

Solution: To simulate data for a regression model where $y = 3x + 2 + \text{noise}$:

```
set.seed(123)
```

```
x <- rnorm(100, mean = 5, sd = 2)
```

```
epsilon <- rnorm(100, mean = 0, sd = 1)
```

```
y <- 3 * x + 2 + epsilon
```

```
lm_model <- lm(y ~ x)
```

```
summary(lm_model)
```

3. Simulation of Data for Time Series Model:

Solution: To simulate a time series from an AR(2) process with AR coefficients 0.5 and -0.3:

```
set.seed(123)
```

```
ar2_process <- arima.sim(n = 200, list(ar = c(0.5, -0.3)))
```

```
plot(ar2_process, main = "Simulated AR(2) Time Series")
```

4. Monte Carlo Simulation:

Solution: To estimate the probability that the sum of two independent uniform random variables exceeds 1:

```
set.seed(123)
```

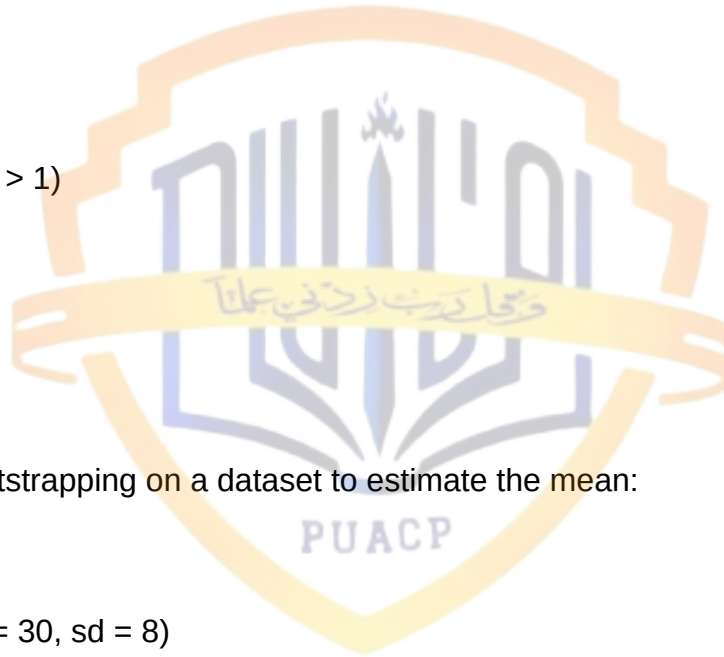
```
n <- 10000
```

```
x <- runif(n)
```

```
y <- runif(n)
```

```
probability <- mean(x + y > 1)
```

```
probability
```



5. Bootstrapping:

Solution: To perform bootstrapping on a dataset to estimate the mean:

```
set.seed(123)
```

```
data <- rnorm(50, mean = 30, sd = 8)
```

```
bootstrap_means <- replicate(1000, mean(sample(data, replace = TRUE)))
```

```
hist(bootstrap_means, main = "Bootstrap Distribution of Means", xlab = "Bootstrap Mean")
```